

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402617
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Hong; Dunhua et al.

Pest trap for tea planting

Abstract

The present disclosure relates to the technical field of pest capturing in protection of tea trees, in particular to a pest trap for tea planting. The pest trap includes a trap framework and a light-emitting diode (LED) lamp; an upper end of the trap framework is provided with a support which is used for clamping the LED lamp; a board for trapping pests coated with sex pheromone on a surface is arranged inside the trap framework; a heating source is arranged on one side of the board for trapping pests and is used for making the sex pheromone to be volatilized faster and farther; a mesh enclosure used for trapping pests is arranged on an outer side of the trap framework; and a small circular hole is formed in a bottom of the mesh enclosure.

Inventors: Hong; Dunhua (Zunyi, CN), Wang; Jiazhen (Zunyi, CN), Song; Dachen (Zunyi, CN), Zou; Ya (Zunyi, CN), Fang; Zhenxing (Zunyi, CN), Zhong; Furu (Zunyi, CN), Jin; Xing (Zunyi, CN), Chen; Hongying (Zunyi, CN), Luo; Yi (Zunyi, CN), Gao; Zhixi (Zunyi, CN), Zhao; Die (Zunyi, CN)

Applicant: ZUNYI NORMAL UNIVERSITY (Zunyi, CN)

Family ID: 87081646

Assignee: ZUNYI NORMAL UNIVERSITY (Zunyi, CN)

Appl. No.: 18/343460

Filed: June 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240381864 A1	Nov. 21, 2024

Foreign Application Priority Data

CN	202310560217.7	May. 17, 2023
----	----------------	---------------

Publication Classification

Int. Cl.: A01M1/04 (20060101); A01M1/02 (20060101); A01M1/10 (20060101)

U.S. Cl.:

CPC A01M1/04 (20130101); A01M1/023 (20130101); A01M1/10 (20130101); A01M1/106 (20130101); A01M2200/01 (20130101)

Field of Classification Search

CPC: A01M (1/00); A01M (1/02); A01M (1/023); A01M (1/026); A01M (1/04); A01M (1/10); A01M (1/106); A01M (1/14); A01M (1/145)

USPC: 43/107; 43/113; 43/114; 43/122

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4168591	12/1978	Shaw	43/118	A01M 1/14
4443964	12/1983	Horn	43/118	A01M 1/02
4654998	12/1986	Clay	43/115	A01M 1/145
5095648	12/1991	Keenan	43/136	A01M 1/02
5184417	12/1992	Weldon	43/107	A01M 1/106
5203816	12/1992	Townsend	43/113	A01M 1/145
5231790	12/1992	Dryden	43/113	A01M 1/145
5513465	12/1995	Demarest	43/113	A01M 1/145
5595018	12/1996	Wilbanks	43/112	A01M 1/223
5799436	12/1997	Nolen	43/112	A01M 1/223
5907923	12/1998	Heath	43/107	A01N 33/04
5915948	12/1998	Kunze	43/113	A01M 1/145
6055766	12/1999	Nolen	43/112	A01M 1/023
6088949	12/1999	Nicosia	43/132.1	A01M 1/023
6199315	12/2000	Suzue	43/113	A01M 1/14
6425202	12/2001	Lin	43/132.1	A01M 1/2016
6516559	12/2002	Simchoni	43/107	A01M 1/023
6655080	12/2002	Spiro	43/107	A01M 1/08
7234268	12/2006	Welch	43/112	A01M 1/02
8109036	12/2011	Wilbanks	43/112	A01M 1/223
10736309	12/2019	Cogley	N/A	H04M 1/72409
11311005	12/2021	Smith	N/A	A01M 1/023
2003/0061757	12/2002	Askin	43/112	A01M 1/023
2003/0070346	12/2002	Winner	43/112	F23D 14/28
2004/0128902	12/2003	Kollars, Jr.	43/107	A01M 1/22
2006/0016120	12/2005	Masters	43/107	A01M 1/14
2009/0313883	12/2008	Olson	424/84	A01M 1/026
2010/0024278	12/2009	Simchoni-Barak	43/107	A01M 1/145

2010/0024279	12/2009	Plato	424/84	A01M 1/2016
2010/0154290	12/2009	Fisher	43/107	A01M 1/10
2012/0124890	12/2011	Hainze	43/121	A01M 1/026
2013/0283671	12/2012	Czokajlo	43/113	A01M 1/026
2020/0352152	12/2019	Stevens	N/A	A01M 1/023

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201294816	12/2008	CN	N/A
204317327	12/2014	CN	N/A
205180143	12/2015	CN	N/A
105532609	12/2015	CN	A01M 1/04
205337306	12/2015	CN	N/A
205492200	12/2015	CN	N/A
206024952	12/2016	CN	N/A
208338723	12/2018	CN	N/A
208597581	12/2018	CN	N/A
209017713	12/2018	CN	N/A
110050772	12/2018	CN	N/A
209300043	12/2018	CN	A01M 1/02
209862041	12/2018	CN	N/A
210382349	12/2019	CN	N/A
210492281	12/2019	CN	N/A
112825828	12/2020	CN	N/A
213344047	12/2020	CN	N/A
213784949	12/2020	CN	N/A
213784950	12/2020	CN	N/A
113367105	12/2020	CN	N/A
214155928	12/2020	CN	N/A
214229552	12/2020	CN	N/A
214546695	12/2020	CN	A01M 1/026
214593780	12/2020	CN	N/A
113812384	12/2020	CN	N/A
214962140	12/2020	CN	N/A
215074920	12/2020	CN	N/A
215074926	12/2020	CN	N/A
215346624	12/2020	CN	N/A
215422388	12/2021	CN	N/A
215531043	12/2021	CN	N/A
215531055	12/2021	CN	N/A
215602749	12/2021	CN	N/A
215992461	12/2021	CN	N/A
216018637	12/2021	CN	N/A
216254856	12/2021	CN	N/A
216821401	12/2021	CN	N/A
216874644	12/2021	CN	N/A
216906537	12/2021	CN	N/A
216961252	12/2021	CN	N/A
217065105	12/2021	CN	N/A
217136541	12/2021	CN	N/A

217242289	12/2021	CN	N/A
217487366	12/2021	CN	N/A
2525897	12/2017	GB	A01M 1/04
100853616	12/2007	KR	N/A
101152776	12/2011	KR	N/A
20150004202	12/2014	KR	N/A
101857372	12/2017	KR	N/A
2105474	12/1997	RU	N/A
WO-2006046067	12/2005	WO	A01M 1/2005
WO-2020081031	12/2019	WO	N/A
WO-2022189845	12/2021	WO	N/A

Primary Examiner: Ark; Darren W

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This patent application claims the benefit and priority of Chinese Patent Application No. 2023105602177, filed with the China National Intellectual Property Administration on May 17, 2023, the disclosure of which is incorporated by reference herein in its entirety as part of the present application.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of pest trapping in protection of tea trees, in particular to a pest trap for tea planting.

BACKGROUND ART

(3) In the agricultural planting industry, especially in a process of tea planting, as production increases, the pests and diseases are becoming more and more serious, causing serious production loss to tea, resulting in a decline in the yield and quality of tea. Based on this, tea farmers often use pesticide to prevent and control pests. Due to application of a large amount of pesticide for a long time, the pesticide resistance of pests in a tea garden increases, so it is not easy to trap and kill pests. Furthermore, there may be pesticide residues that affect the quality and safety of tea, which is not conducive to people's health. Excessive pesticide will also cause a harmful effect to the ecological environment.

SUMMARY

(4) The present disclosure aims to provide a pest trap for tea planting, which can achieve a better trapping effect on pests in a tea garden and is more convenient to use and pollution-free to the environment.

(5) The present disclosure provides a pest trap for tea planting, including a trap framework and a light-emitting diode (LED) lamp; an upper end of the trap framework is provided with a support which is used for clamping the LED lamp; a board for trapping pests coated with sex pheromone on a surface is arranged inside the trap framework; a heating source is arranged on one side of the board for trapping pests and is used for volatilizing the sex pheromone faster and farther; a mesh enclosure used for trapping pests is arranged on an outer side of the trap framework; and a small circular hole is formed in a bottom of the mesh enclosure.

(6) Further, the sex pheromone is female sex pheromone. Corresponding female sex pheromone is selected according to the species of a pest to be prevented and is coated on the board for trapping pests to achieve an effect of trapping male pests.

(7) Further, a bottom of the trap framework is provided with supporting legs.

(8) Further, the trap framework is cylindrical.

(9) Further, the board for trapping pests is a hollow cylinder formed by splicing a blue rectangular board and a yellow rectangular board, and is placed at a center position inside the trap framework. Due to certain sensitivity to some colors, insects often tend to fly to objects in these colors. The insects have good color tropism to yellow color and blue color. According to the color tropism of insects, the yellow and blue board for trapping pests is used to trap pests into the trap, making the pests fail to fly, thus achieving the purpose of preventing and treating pests.

(10) Further, a surface of the board for trapping pests is a smooth material since a non-viscous board for trapping pests can be cyclically and repeatedly used for a long time.

(11) Further, the yellow has a wavelength of $575\text{ nm}\pm 10\text{ nm}$, and the blue has a wavelength of $465\text{ nm}\pm 10\text{ nm}$.

(12) Further, the female sex pheromone is sprayed to an inner surface of the hollow cylinder, and the heating source is placed on the outer side of the hollow cylinder to enable the sex pheromone inside the board for trapping pests to achieve the best volatilization effect.

(13) Further, the heating source is a semiconductor heating sheet which can achieve a heating effect on a space, is safe to use, and has the advantages of insulation, pressure resistance, flame resistance and the like. By means of heating the female sex pheromone on the board for trapping pests, the female sex pheromone can be volatilized faster and farther, thus trapping more pests.

(14) Further, a top of the trap framework is provided with a rain cover which can protect the trap from being exposed to the sun and rain.

Beneficial Effects

(15) According to the pest trap, the LED lamp emits light for stimulation. A living being has tropism to the stimulation of light, that is, insects have phototaxis and will fly to a light source. Due to the phototaxis, the insects are trapped into the trap, thus achieving an insect expelling effect. The sex pheromone is one kind of insect pheromone, which achieves communication between insect individuals of the same species for courting, foraging, perching and other activities. The insect pheromone is secreted and released by insects in the bodies. The released pheromone can allure heterosexual individuals and can cause the heterosexual individuals to arise reproductive reactions for foraging and directional mating. Synthetic female sex pheromone is released in a tea garden to trap male pests into the pest trap, so that the male pests and female pests are prevented from mating, thus reducing the number of pests and also reducing pests of the next generation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe the specific implementation modes of the present disclosure or the technical solutions in the prior art more clearly, drawings required to be used in the specific implementation modes or the illustration of the prior art will be briefly introduced below. Apparently, the drawings in the illustration below are some implementation modes of the present disclosure. Those ordinarily skilled in the art can also obtain other drawings according to these drawings without doing creative work.

(2) FIG. 1 is an overall schematic diagram of the present disclosure;

(3) FIG. 2 is a schematic diagram of a trap framework and an inside of the trap framework of the present disclosure;

(4) FIG. 3 is a schematic diagram of a bottom of the present disclosure.

(5) Reference signs in the drawings: **1**: trap framework; **2**: LED lamp; **3**: support; **4**: board for trapping pests; **5**: mesh enclosure; **6**: small circular hole; **7**: supporting leg; **8**: semiconductor heating sheet; **9**: rain cover; and **10**: bottom plate.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(6) Technical solutions of the present disclosure will be described clearly and completely below in combination with the embodiments. Obviously, the embodiments described herein are part of the embodiments of the present disclosure, not all the embodiments. Based on the embodiments in the present disclosure, all other embodiments obtained by those of ordinary skill in the art without creative work shall fall within the protection scope of the present disclosure.

(7) In the description of the present disclosure, it should be understood that orientations or positional relationships indicated by the terms “center”, “longitudinal”, “transverse”, “length” “width”, “thickness”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inside”, “outside”, “clockwise”, “anticlockwise”, and the like are orientations or positional relationships as shown in the drawings, and are only for the purpose of facilitating and simplifying the description of the present disclosure instead of indicating or implying that devices or elements indicated must have particular orientations, and be constructed and operated in the particular orientations, so that these terms are not construed as limiting the present disclosure.

(8) In addition, the terms “first” and “second” are used for descriptive purposes only and are not to be understood to indicate or imply relative importance or to imply the number of indicated technical features. Therefore, features defined by “first” and “second” can explicitly instruct or impliedly include one or more features. In the description of the present disclosure, unless expressly specified otherwise, the meaning of the “plurality” is two or more than two. In addition, the terms “mount”, “connect” and “connection” shall be understood broadly. For example, it may be fixed connection, detachable connection, integral connection, mechanical connection, electrical connection, direct connection, indirect connection via an intermediate, or inter-connection between two elements. Those of ordinary skill in the art can understand the specific meanings of the above terms in the present disclosure according to specific situations.

Embodiment 1

(9) A pest trap for tea planting, as shown in FIGS. 1-3, includes a cylindrical trap framework **1** and an LED lamp **2**. A bottom of the trap framework **1** is provided with supporting legs **7**; and an upper end of the trap framework **1** is provided with a support **3** which is used for clamping the LED lamp **2**. A blue and yellow splice of hollow cylindrical trap board **4** surrounds the inner center of trap frame **1**; a bottom of the trap framework **1** is provided with a bottom plate **10**; a circular hole is formed in the middle of the bottom plate **10**; a diameter of the circular hole is slightly less than that of the hollow cylindrical board **4** for trapping pests; the hollow cylindrical board **4** for trapping pests is erected above the circular hole, and an inner surface of the hollow cylindrical board **4** for trapping pests is coated with female sex pheromone (the sex pheromone on the board **4** for trapping pests is pest sex pheromone, aggregation pheromone or plant meta-pheromone for Empoascavitis, aleyrodid, thrips and the like in a tea garden); a semiconductor heating sheet **8** is attached to an outer side of the hollow cylindrical board **4** for trapping pests; and the semiconductor heating sheet **8** can achieve a heating effect on a space, is safe to use and has the advantages of insulation, pressure resistance, flame resistance and the like. By means of heating the female sex pheromone on the board **4** for trapping pests, the female sex pheromone can be volatilized faster and farther, thus trapping more pests. A mesh enclosure **5** used for trapping pests is arranged on the outer side of the trap framework **1**. A small circular hole **6** is formed in a bottom of the mesh enclosure **5**. The small circular hole **6** is a passage for allowing pests to get into the trap. The yellow of the board for trapping pests has a wavelength of $575\text{ nm}\pm 10\text{ nm}$, and the blue has a wavelength of $465\text{ nm}\pm 10\text{ nm}$; the yellow board has a wider pest trapping area, and the blue board is more targeted in pest killing. The two kinds of boards for trapping pests are combined for use to greatly enhance the pest trapping effect. There are many kinds of pests in the tea garden, and different pests have different tendency to colors. For example, Empoascavitis has tropism to yellow color, and thrips has tropism to blue color. In the pest prevention and treatment process, there is a trouble since insects have different color tropisms. It is more efficient to splice the yellow and blue boards for trapping pests into the hollow cylindrical board for trapping pests. A top of the trap framework **1** is also provided

with a rain cover 9, which can protect the trap from being exposed to the sun and rain, so that the service life of the trap is prolonged.

(10) Working Principle:

(11) The present disclosure provides a trap applied to a tea garden to trap small pests such as Empoascavitis, tea geometrid, and thrips. The hollow cylindrical board for trapping pests formed by splicing the yellow and blue boards for trapping pests is encircled in the center of the trap; the semiconductor heating sheet is attached to the outer side of the hollow cylindrical board for trapping pests; the inner surface of the hollow cylindrical board for trapping pests is sprayed with the sex pheromone; after the heating sheet generates heat, the sex pheromone can be catalyzed to attract pests to the maximum extent; meanwhile, the colored hollow cylindrical board for trapping pests can also attract pests; and in addition, pests need to actively tend to a room-temperature place in the environment in order to survive, that is, the pests have thermotaxis. The LED lamp emits light and heat, and the semiconductor heating sheet generates heat, so that pests can be effectively trapped. By combining the four components, pests are trapped by fully using color, sex pheromone, light and temperature. The pests are induced to pass through the small circular hole and get into the trap, so that the pest trapping rate is greatly increased. A circle of fine transparent mesh enclosure is covered outside the trap, which can ensure that the pests that have entered the trap cannot fly out. A rainproof device is placed at the top end of the trap, which can protect the trap from being exposed to the sun and rain. When the trap is applied to a tea garden to trap pests, the trap is placed between two tea groups. After the prevention and treatment end, trapped pests can be used as fertilizer for the tea garden. The trap is recyclable.

(12) It should be finally noted that the above various embodiments are only used to describe the technical solutions of the utility model, and not intended to limit the utility model. Although the utility model has been described in detail with reference to the foregoing embodiments, those ordinarily skilled in the art should understand that they can still modify the technical solutions described in all the foregoing embodiments, or equivalently replace some or all of the technical features, and these modifications or replacements do not depart the essences of the corresponding technical solutions from the spirit and scope of the technical solutions of all the embodiments of the utility model.

Claims

1. A pest trap for tea planting comprising: a trap framework and a light-emitting diode lamp, wherein an upper end of the trap framework is provided with a support which is used for clamping the light-emitting diode lamp; a board for trapping pests being coated with a female sex pheromone on a surface thereof and being arranged inside the trap framework, wherein the board for trapping pests is a hollow cylinder formed by splicing a blue rectangular board and a yellow rectangular board and is placed at a center position inside the trap framework, wherein the surface of the board for trapping pests is a smooth material, wherein the yellow has a wavelength of $575\text{ nm}\pm 10\text{ nm}$, and the blue has a wavelength of $465\text{ nm}\pm 10\text{ nm}$, and wherein the female sex pheromone is sprayed to an inner surface of the hollow cylinder; a heating source is arranged on one side of the board for trapping pests and is used for making the female sex pheromone to be volatilized faster and farther, wherein the heating source is placed on an outer side of the hollow cylinder; a mesh enclosure used for trapping pests is arranged on an outer side of the trap framework; and a small circular hole is formed in a bottom of the mesh enclosure.
2. The pest trap for tea planting according to claim 1, wherein a bottom of the trap framework is provided with supporting legs.
3. The pest trap for tea planting according to claim 1, wherein the trap framework is cylindrical.
4. The pest trap for tea planting according to claim 1, wherein the heating source is a semiconductor heating sheet.

5. The pest trap for tea planting according to claim 1, wherein a top of the trap framework is provided with a rain cover.
